

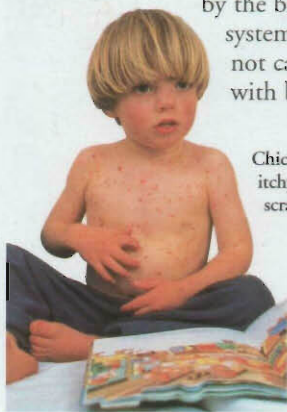
DISEASES



JUST LIKE A MACHINE, the human body works smoothly and efficiently most of the time. However, it may occasionally stop operating normally. This may be due to an injury, such as a broken bone, but, more commonly, it is caused by a disease. Diseases occur because the body has been infected by a pathogen (germ), as in the case of influenza or food poisoning, or because of problems arising inside the body, such as heart disease or diabetes. Some diseases can be controlled and defeated by the body's immune system. More serious diseases may need drug treatment or surgery in order to cure them.

Infectious diseases

Infectious diseases are those, such as the common cold or pneumonia, that are caused by pathogens that invade the body. The most common pathogens are bacteria and viruses, although some diseases, such as thrush, are caused by fungi, and some, such as malaria, by tiny organisms called protists. They are normally destroyed by the body's immune system. Those that are not can often be dealt with by drugs.



Chickenpox causes an itchy rash that, when scratched, can leave scars.



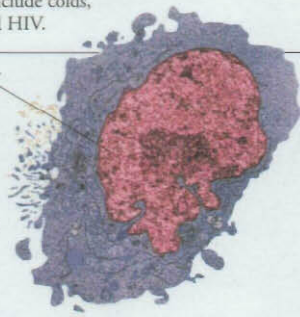
Bacteria are in water, air, and soil, as well as many plant and animal tissues.

Bacteria

Bacteria are single-celled micro-organisms. Most bacteria are not harmful to humans. However, some multiply inside the body and produce toxins that cause disease. Bacterial diseases include typhoid and scarlet fever. Most can be treated with drugs called antibiotics.

Viruses

Viruses are tiny infective particles, not usually classed as living things. They take over a body cell's genetic material (DNA) and make copies of themselves that infect other cells. Human viral infections include colds, measles, and HIV.



HIV and AIDS

The human immunodeficiency virus, or HIV, causes AIDS (Acquired Immune Deficiency Syndrome). HIV infects and destroys the cells that form part of the body's immune system – the body's defences against diseases. HIV is transmitted by some bodily fluids, such as blood and semen. The system becomes progressively weaker, and the person becomes infected with various diseases, known collectively as AIDS.

Preventing disease

Disease prevention is an important part of modern medicine. Diseases can be prevented by better sanitation, immunization, and improving food hygiene. Eating a balanced diet and exercising may also prevent disease.



Syringes that are not properly sterilized after use can spread disease.



Epidemiologist tests samples in laboratory.

Epidemiology

Epidemiology is the study of diseases as they affect groups of people. Epidemiologists are concerned with why diseases occur in a population, and their control and prevention. They have discovered links between disease and diet, environmental factors, and lifestyle. Epidemiologists first discovered the link between smoking and lung cancer.

Non-infectious diseases

If a disease is non-infectious, it is not caused by a pathogen and cannot be passed from one person to another. Non-infectious diseases include circulatory system diseases, such as heart attacks, strokes, and cancer, and respiratory diseases, such as bronchitis and emphysema.



Nutritional diseases

Nutritional diseases are caused by a lack of a balanced diet, causing a deficiency of vitamins and minerals. A child not getting enough vitamin D may suffer from rickets, where the skeleton does not form properly.

Miners may develop lung problems.



Industrial diseases

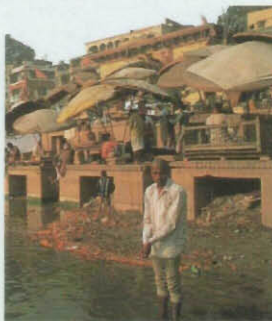
Work situations may affect a person's health. Industrial processes can create harmful environments or use chemicals that cause diseases. Some miners develop a lung disease called pneumoconiosis.

Spreading infection

Most diseases are acquired from other people by skin-to-skin contact, breathing in droplets when someone sneezes or coughs, or by sexual contact without the use of condoms. Infection can also be spread through infected food, contaminated water, and insect bites. Drug users who share needles risk infections of the blood, such as hepatitis and HIV.

Sanitation

Human faeces contain bacteria and viruses that cause disease. If there is poor sanitation and human waste is discharged into rivers, people may catch diseases such as dysentery or cholera through contact with polluted water.



Some mosquitoes carry strains of malaria that are resistant to drugs.

Insects

Insects such as mosquitoes and fleas feed on human blood and can carry disease. A mosquito transmits the malaria micro-organism if it feeds on an infected person's blood.

Keeping rivers clean prevents diseases that can be caught if people drink, wash, or grow food in the water.



Bottled water is a way of assuring water is safe in certain countries.



Pills contain measured amounts of drugs.